No of additional sheets 1

LOURDES MATHA COLLEGE OF SCIENCE & TECHNOLOGY

LOURDES HILLS, KUTTICHAL P. O., THIRUVANANTHAPURAM- 695574

ANSWER BOOK

Name Febin T Rajeev Class No 20 Branch MCA

Subject Deep Learning Semester Third

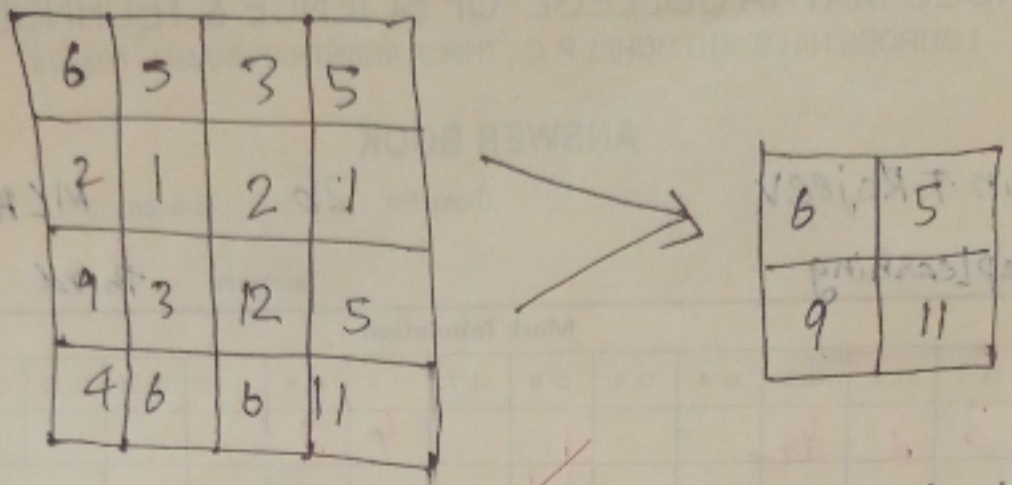
Mark Tabulation

		Q.1	Q.2	Q.3	Q.4	Q.5	Q.6	Q.7	Q.8	Q.9	Q.10	Q.11	Q.12	Q.13	Q.14	Q.15
Mark Divisions		3	2	½			1		4	2						
Total																
Grand Total														13		

1) Max Pooling in the context of CNN

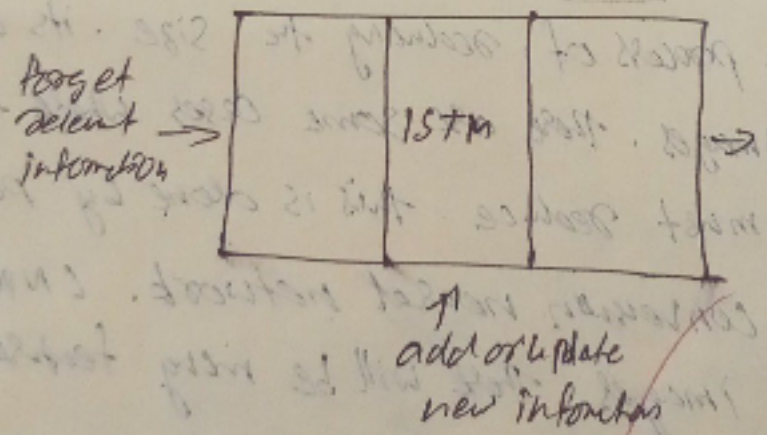
Pooling is the process of reducing the size. its done mainly in images. there are some cases where the image size must reduce. this is done by pooling. CNN means convolution neural network. CNN used for classifing images. here will be many features like objects colours each so on. these all are classified using CNN. max pooling is the process of taking the maximum among the systems or max count in the table / matrix.

for example.



this is how max pooling is done in the context of CNN.

8) LSTM stands for long short term memory It an advanced RNN. Its also has 3 major main parts they are



The working of LSTM is firstly the previous time stamp has some data / information the system can either remember the information or forget the information.

and the secondly they tries to get information to the System. this LSTM has ~~the~~ ~~three~~ three gates first one is forget gate, secondly input gate, thirdly output gate.

LSTM has a cell state where core state goes on. this is represented as C(T-1)

The main use of LSTM is that it is used to solve vanishing gradient problem. Its one of the main merit of ~~GRU~~ LSTM - LSTM is one of the main advanced technology used in ANN. The main thing is that the information in the ~~next~~ previous depent on it (LSTM). This LSTM only takes the relevant information for the working and ignores the irrelevant informations.

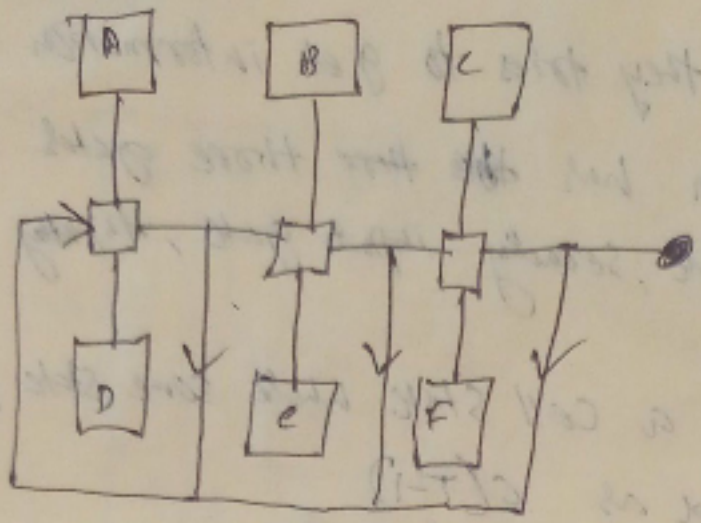
This is how LSTM works.

4

9)

truncated back propagation through time

Its used for training the ANN. Here the difficult is that back propagation is applied to every time stamp. thats why its called back propagation through time.



The training is done by this manner. Which means the training is done with a back propagation.

2 The back propagation means its goes back for manipulating the values and again sending the input that how back propagation is done. And its done in every time interval / time stamp.

By doing this the output / the value's accuracy will be straight. This is one of main use and benifit of ~~back~~ trained back propagation.

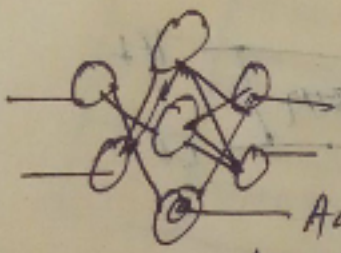
3) The problem of vanishing gradient ~~with~~ can be solved by LSTM / Long short term memory.

Its has forgetting gate, its used for remembering the informations. another one is input gate the all input values of vanishing gradient is stored in that gate. And the final gate is that output gate its done to get the output of the vanishing gradient.

By using LSTM some patterns / hidden details which is hidden in a image can be detected using LSTM.

2) ReLU

ReLU is an important activation function used in ANN.



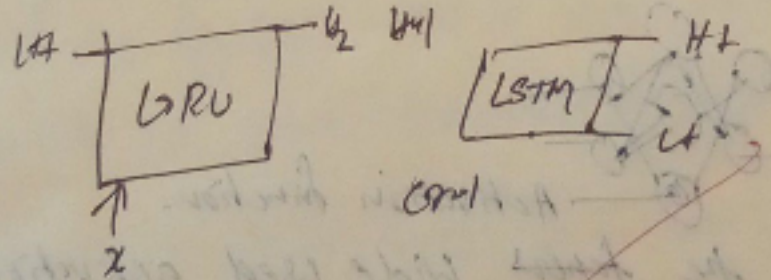
Activation function.

Relus is the ~~famous~~ wide used activation function. it takes some data as input and provides an output. the best data will be selected using these activation function. its Advantage which has some polynom equations and mathematical functions included.

the relu is also used in car (automated cars) like tesla etc.
its complexity is very less compared to others.
the activation function is at ~~negative~~ positive then the
modified value is passed or if it is negative value
the same value is passed.

2 A neuron is activated when the activation function
is activated. It removes the negative grayscale
values and passes only positive ones.
the output of activation function is in between
0 and input value

6) (GRUs) or Gated Recurrent units is similar to
LSTM and it uses gate to control the flow of
information.



Its faster than LSTM. But its does not have
Cell State